

Economic Report of the Group

Overall Economic and Industry Conditions

The German economy continues to be affected by the COVID-19 pandemic¹⁵

The COVID-19 pandemic continues to shape the economic situation in Germany. After experiencing a slowdown in the winter half-year 2020/21 due to the pandemic, the German economy continued to recover from the second quarter of 2021 onwards. According to calculations published by the Federal Statistical Office (Destatis), gross domestic product (GDP) grew by 2.0% (calendar-adjusted) in the second quarter of 2021 compared to the same period in the prior year. In the third quarter of 2021, GDP increased by 1.7% compared to the second quarter of 2021 – adjusted for price, seasonal and calendar effects. The decline in new infections in the summer months made possible widespread opening steps in previously restricted sectors of the economy. On the demand side, private consumption was the main growth driver for the summer recovery. Industrial production also continued to decline. The primary reasons behind this were persistent supply bottlenecks and shortages of primary products and raw materials.

Economists at the Bundesbank forecast that the growth of the German economy will have slowed by year-end 2021 due to supply bottlenecks and the worsening of the COVID-19 infection situation. For 2021 as a whole, experts at the Federal Statistical Office estimate real gross domestic product growth of approximately 2.8% (calendar-adjusted).

Inflation growth has accelerated since the beginning of 2021, according to data from the Federal Statistical Office (Destatis).

There are several reasons for this, including base effects from low prices in 2020. In particular, the temporary reduction in VAT in the second half of 2020 and the drop in the price of mineral oil products in 2020 are having an increasing effect. Other factors include the introduction of CO₂ pricing since January 2021 and factors related to the crisis, such as price increases at the upstream economic stages. Preliminary estimates by the Federal Statistical Office estimate that the inflation rate in Germany was +5.3% in December 2021 and +5.2% in November 2021 compared to the same month of the previous year. Energy, up 18.3% year-on-year, was again the main price driver in December. Food prices rose by 6.0%, services by 3.1%.

After the easing of pandemic measures in the spring of 2021, the labour market recovered and improved in 2021 but then slowed again in the autumn of 2021. Figures from the Federal Employment Agency indicate that the number of unemployed fell from 2.7 million at the end of 2020 to 2.3 million in December 2021. The unemployment rate fell from 5.9% to 5.1% over this period.

Technology trends bring growth potential for the telecommunications market¹⁶

The COVID-19 crisis has clearly shown the important role of telecommunications in our society and our economy. The pandemic has changed our everyday lives and the world of work. The use of digital solutions for work, leisure and shopping have increased further and has accelerated a number of (digital) developments. Connectivity in particular has proven to be an important component of everyday life. In particular due to working from home, home schooling, but also to entertainment services at home, consumers' demand for internet access with ever higher bandwidth is increasing.

¹⁵ Sources: German Federal Statistical Office (Destatis): Press releases № 532 (25 November 2021), № 541 (29 November 2021), № 005 (6 January 2022) and № 039 (28 January 2022); Bundesbank: Monthly Report November (22 November 2021) and December (20 December 2021); Bundesbank: Press release "Bundesbank projections: Upswing shifts somewhat" (17 December 2021); Federal Employment Agency (Bundesanstalt für Arbeit): Monthly Report December 2021 (4 January 2022); German Council of Economic Experts: Annual expert reports 2021/22 (10 November 2021)

¹⁶ Sources: VATM: "Telecommunications Market Study 2021" (28 October 2021); Deutsche Telekom AG: Press release 1st half of 2021/22 (16 November 2021); Digital association Bitkom: Future of Consumer Technology Study 2021 (22 September 2021); Ad Alliance: Press release "Mobile 360° Study: Smartphones are now essential" (18 November 2021); PC Welt: 5G networks increasingly gaining traction (6 August 2021); Deloitte and BVDW: Study "Consumer IoT Fact Check" (10 March 2021); mm1 Consulting: Study "10 Years of Fixed-Mobile Convergence" (June 2020); Statista: Global Consumer Survey (September 2021); PC Welt: Mobile routers under test (27 October 2021)

Fast internet is also provided by 5G technology, which is available in more and more areas of Germany. 5G offers a wide range of application possibilities in both the private and business sectors. This will include, for example, offerings such as 4K-/8K video, virtual and augmented reality and real-time gaming, as well as connected and autonomous cars. The growing range of high-end, mid-range and entry-level 5G-enabled smartphones is also driving the adoption of the new mobile communications standard. VATM estimates that by the end of 2021, 10.8 million SIM cards will allow the use of fifth-generation mobile networks.

Since the beginning of the coronavirus crisis, the Germans have been using their smartphones more than before. According to a study by Ad Alliance, the smartphone is now the preferred means of communication for many people, and at the same time it is a source of knowledge, an assistant in everyday life, an entertainer, a games console, a fitness partner and a wallet. The use of apps has increased correspondingly, especially in the areas of health, gaming, shopping and delivery services. In addition, apps that simplify the transition to the digital and contactless payment world are also becoming increasingly important in the market. A quarter of the people surveyed already use the option of mobile payment for stationary retail. According to another Bitkom survey, the smartphone became the most popular online shopping device in 2021. Today, 6 out of 10 people use their smartphone to make online purchases. In the previous year, the laptop was still in first place.

LTE and 5G routers, are becoming increasingly popular as an alternative to fixed broadband connections for accessing the internet. The advantage over cable or DSL is that the devices can be used immediately, without the need for a technician. The internet connection runs via the mobile network. Depending on the provider or the product, these routers can be used at specific locations or independent of location. Typical use cases are, for example, in a second home, in a camper or at a campsite. But even in your own home, you can use a HomeSpot or LTE or 5G router as a replacement for classic fixed-line connections via DSL or cable, or as a bridging solution for Wifi while, for example, the fibre-optic network is being expanded in your neighbourhood. A survey by Statista found that around 11% of respondents now access the internet at home via mobile internet routers.

Telephony continued to be very popular in 2021, according to VATM data. Due to the pandemic, the number of fixed-line minutes also increased in 2020 for the first time in 13 years – and even in 2021, fixed-line calls were still made slightly more frequently, especially during lockdown periods. People are reaching for their smartphones more frequently and overall for longer periods of time: It is estimated that 433 million minutes a day were spent talking on mobile phones in 2021 and 234 million minutes were spent using over-the-top (OTT) providers.

The Internet of Things (IoT) and the smart connection of items play an important role not only in the transformation of

companies and infrastructures: They have long since become part of consumers' everyday lives. According to a study by Deloitte and BVDW, seven out of ten consumers in Germany currently use at least one end device from the world of the Internet of Things. The products used range from connected TVs and game consoles to smart speakers and smartphone voice assistants to wearables such as smartwatches and fitness trackers for athletes. Smart homes also feature a wide range of devices. In particular, voice assistants are increasingly being used alongside smartphones as control centres, either in the form of separate smart speakers or as integrated voice control software. A study by Bitkom revealed that 44% of all internet users from 16 years on use voice commands to request information or control devices, at least from time to time, across all devices. That is 5 percentage points more than in the previous year. The smartphone is the leading device when it comes to the use of digital voice assistants.

The television market in Germany is also in a state of upheaval. This development was accelerated by the pandemic as well. The consumption of films, series and video clips via the internet was already an integral part of everyday media use in Germany before the crisis and has become even more important, especially during the lockdown. Video-on-demand (VoD) services are distributed over the top (OTT), i.e. via the Internet. Traditional linear television is in competition with these services, especially among the young target group. The continuous increase in the consumption of VoD content is supported by the flexible use of devices such as tablets or smartphones. Although linear television is losing overall popularity, the COVID-19 crisis has highlighted its importance as a reliable source of information. For example, according to a study by the digital association Bitkom, the smartphone is now the most popular device for daily video streaming: almost a third of those who stream use it every day to watch videos, a fifth of users do so multiple times each week.

Demand for mobile data services continues to grow¹⁷

The mobile phone market is a saturated market. According to VATM data, the number of SIM cards on the market at the end of 2021, including M2M or IoT cards, was 157.8 million (end of 2020: 150.0 million). The strong SIM card growth is mainly due to M2M and IoT cards: According to VATM, at the end of 2021, around 40.2 million SIM cards were in use for M2M and IoT applications, compared to 36.1 million at the end of 2020. These M2M or IoT cards accounted for over 25% of all SIM cards on the market by the end of 2021.

The continued demand for more data-intensive Internet services such as video streaming and social media was responsible for a further increase in mobile data use in 2021. According to VATM estimates, the average monthly data consumption per mobile phone customer increased from 2.6 GB in 2020 to 3.3 GB in 2021. This corresponds to an increase of 26%.

¹⁷Sources: VATM: "Telecommunications Market Study 2021" (28 October 2021); Company data and own calculations: Deutsche Telekom Investor Relations: publication of Q3 2021 results (12 November 2021); Vodafone Deutschland press release July-September 2021 results (16 November 2021); 1&1 AG: Interim Report Q3 2021 (9 November 2021)

Telefónica Deutschland Group continues to gain service revenue market share in the German mobile network operator market¹⁸

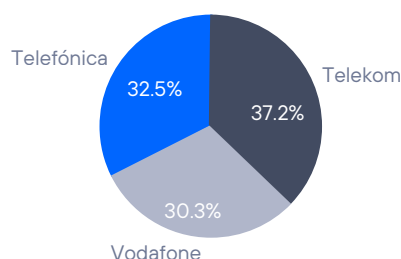
The German mobile telecommunications market currently consists of three network operators and several service providers and mobile virtual network operators (MVNOs). At the 5G frequency auction, which ended in June 2019, the provider 1&1 acquired frequencies and plans to establish itself as the fourth wireless services network operator in the future and build its own mobile network. The 1&1 Group concluded a long-term national roaming agreement with Telefónica Deutschland Group in May 2021, which will provide to 1&1 customers nationwide mobile coverage in the areas not yet served by the 1&1 Group during their 5G network roll-out phase.

In 2020, the mobile service revenues in the German mobile market were negatively impacted by the pandemic, for example through negative effects on roaming revenues, as well as through shop closures. However, the market largely recovered in 2021. The market grew by 2.0% year-on-year in the nine-month period from January to September 2021. Telefónica Deutschland Group increased its market share in mobile network operator service revenue in the nine months from January to September from 31.9% in 2020 to 32.5% in 2021, making it the number two player in the German market.

G 04

MARKET SHARE IN THE MOBILE COMMUNICATIONS MARKET

based on service revenue of network operators (in %)
Q1 to Q3 2021



Growth in German fixed line broadband market continues¹⁹

Growth in fixed line broadband connections continued unabated: VATM estimates that the number of accesses at the end of 2021 increased by around 3% year-on-year to approx. 37.4 million. DSL continues to be the dominant technology, with a share of approx. 69% of fixed-line broadband connections. With 9.0 million active broadband connections at the end of 2021, cable now accounts for a share of around 24% of the total market. "Real" fibre optic connections (FTTH/FTTB) continue to gain importance: According to VATM 2021, they have increased by more than 30% compared to the previous year and now account for almost 7% of active broadband connections. The Telefónica Deutschland Group relies on partnerships in its fixed line business. The Group reaches around 34 million households with VDSL through Deutsche Telekom's infrastructure and, in the future, even more households will be reached with optical fibre (FTTH). As part of the cooperation with Vodafone, the Telefónica Deutschland Group has been able supply up to 24 million cable households in Germany with fixed-line products largely with higher download speeds than VDSL since January 2021. In addition, the cooperation with Tele Columbus has expanded service to another 2.4 million households since July 2021. The Telefónica Deutschland Group connects more households with advanced FTTH connectivity through a partnership with Unsere Grüne Glasfaser (UGG), which was established in 2020 as an independent wholesale company by Telefónica, S.A. Group / Telefónica Deutschland Group and Allianz Group, with Telefónica Deutschland Group holding a 10% stake. Overall, the trend towards higher-speed connections, driven by the COVID-19 pandemic and the need for working from home and home schooling, continues in Germany. According to VATM estimates, more than 52% of customers with a broadband connection will be using maximum download data rates of more than 50 Mbit/s by the end of 2021, compared to 49% at the end of 2020. The increased customer demand for more bandwidth is also reflected in the data volume generated per broadband connection and month. According to VATM, this increased by 30% in 2021 compared to 2020, to an average of 231 GB per access.

¹⁸ Sources: VATM: "Telecommunications Market Study 2021" (28 October 2021); Company data and own calculations: Deutsche Telekom AG Investor Relations: publication of 2020 "Back Up" annual results and 2020 Annual Report (26 February 2021) and Q3 2021 results (12 November 2021); Vodafone Investor Relations Vodafone Group: July-September 2020 results "Vodafone FY21 H1 results announcement" (16 November 2020); July-September 2021 results "Excel Spreadsheet" and "Vodafone FY22 H1 results announcement" (16 November 2021); 1&1 AG: Interim Report Q3 2021 (9 November 2021)

¹⁹ Sources: VATM: "Telecommunications Market Study 2021" (28 October 2021); Deutsche Telekom AG press release: "Broadband expansion: 174,000 households now have faster internet access with Telekom" (27 September 2021); Telefónica Deutschland Holding AG press release: "O₂ launches nationwide roll-out of cable connections" (26 January 2021); Tele Columbus press release: "Tele Columbus launches marketing activities with first wholesale partner" (14 July 2021); Telefónica Deutschland Holding AG press release: "O₂ relies on strong fibre-optic network coverage in Germany for its customers" (2 March 2021)

Regulatory Influences on the Telefónica Deutschland Group

As a provider of telecommunications services and an operator of telecommunications networks, the Telefónica Deutschland Group is required to meet certain regulatory requirements. As such, it is subject to supervision by the Bundesnetzagentur (BNetzA – German Federal Network Agency).

The key regulatory events affecting the Telefónica Deutschland Group in the year under review are discussed below.

Frequencies

BNetzA continues to provide mobile radio frequencies with a scenarios paper

Following the "Spectrum Compass 2020" published for consultation in August 2020, the BNetzA continued the procedure with publication of the "Principles and Scenarios for the Provision of the 800 MHz, 1.8 GHz and 2.6 GHz Spectrum" (scenarios paper) on 21 June 2021, with the aim of exploring the next steps in providing the frequencies. This document describes five scenarios for public consultation, relating to future provision of the 800 MHz usage rights in particular (auction, 800 MHz extension, 800 MHz operator model, combination of auction and extension elements, invitation to tender). The Telefónica Deutschland Group commented in due time and advocated for an extension of the frequencies to be made available through at least 2033. On the basis of the statements submitted concerning the scenario paper, the BNetzA has developed landmarks and put them up for consultation by 21 March 2022. This is accompanied by an initial demand survey to investigate indications of a possible shortage of frequencies. The landmarks essentially only mention an extension and/or auction as the most likely provision mechanisms at the moment.

2019 Frequency Auction

The appeal in the action brought by the Telefónica Deutschland Group against the decisions issued on 26 November 2018 by the Presidential Chamber of the BNetzA (Presidential Chamber Decisions III and IV) about the rules governing the award and the auction of frequencies in the 2 GHz and 3.6 GHz ranges was heard orally before the Federal Administrative Court on 20 October 2021. The Federal Administrative Court dismissed the action in the last instance in a ruling handed down on the same day. This concluded the legal action.

Coverage requirements resulting from the 2019 frequency auction and improved network coverage in "grey spots"

In July 2021, the Telefónica Deutschland Group, the Deutsche Telekom Group and the Vodafone Group entered into a cooperation for the joint construction of additional radio towers and masts, as well as their technical support and use. The cooperation is designed to meet coverage obligations, especially for transport routes and in rural areas. The plan is for the participating companies to construct an equal number of new sites, which can then be used by the cooperation partners as needed on equal terms and equipped with their own antennas and network technology. The 2019 spectrum auction's coverage requirements include far-reaching obligations for

transport routes on railways, federal highways and rural or state roads, for so-called "white spots", as well as for seaports and waterways. Frequency holders are allowed to enter into cooperation agreements to fulfil these obligations.

In November 2021, the Telefónica Deutschland Group concluded an agreement with the Deutsche Telekom Group on active shared network usage at "grey spots", which are areas in which only one mobile network operator offers mobile network access to its customers. A similar agreement with the Vodafone Group has been concluded on 25 January 2022.

Coverage requirements resulting from the 2015 frequency auction

In March 2021, the BNetzA confirmed that the Telefónica Deutschland Group has met the coverage requirements in relation to the households nationwide and in every federal state. With regard to the main traffic routes, the BNetzA confirmed in December 2021 that the Telefónica Deutschland Group had also complied with this coverage requirement. This means that the Telefónica Deutschland Group has fulfilled all coverage requirements from the 2015 frequency auction.

Telecommunications market

Conditions from the Merger of the Telefónica Deutschland Group and E-Plus

The Telefónica Deutschland Group entered into a National Roaming Agreement (NRA) with the 1&1 Group (formerly: 1&1 Drillisch Group) on 21 May 2021. With this NRA, the Telefónica Deutschland Group fulfils another EU requirement imposed by the EU Commission arising from the merger of the Telefónica Deutschland Group and E-Plus in 2014.

On 22 February 2019, the EU Commission opened formal proceedings against the Telefónica Deutschland Group by submitting its grounds for objection arising from the implementation of 4G wholesale access as a result of the merger between the Telefónica Deutschland Group and E-Plus. On 12 March 2021, the EU Commission discontinued the proceedings. A fine was not imposed.

New Telecommunications Act for the Implementation of the EU Electronic Communications Code entered into force on 1 December 2021

The amended Telecommunications Act ("TKG") revises the regulatory framework in the telecommunications sector. This will also result in significant changes for the Telefónica Deutschland Group in many areas. The legal precedence with regard to the provision of frequencies in the event of spectrum scarcity has been dropped in favour of an auction, and extensions and invitations to tender for frequency allocations have also become possible in this case. The amended Telecommunications Act also contains extensive changes in the "Customer Protection" section. Among other things, the obligation to provide information and the right to terminate the contract have been expanded, meaning that a contract that has been tacitly renewed after the expiry of the minimum term can be terminated at any time with a notice period of one month. In addition, both mobile and fixed-line customers will have a special right of termination and a right to a reduction in price if the service provided deviates

from the agreed service. The BNetzA has published a General Administrative Order for fixed-network internet access with effect from 13 December 2021 to specify the vague legal terms "significant, continuous or regularly recurring deviation in speed" and guidelines on a monitoring mechanism for furnishing proof of such deviations. In the mobile sector, the details are still being worked out by the BNetzA. The new Telecommunications Act also adapts and expands the powers and tasks of the BNetzA, such as powers with regard to broadband cooperation, the structuring of universal service and expanded information rights regarding network expansion. Among other effects, the regulations on the publication of financial statements are changing, which will require additional companies of the Telefónica Deutschland Group to publish their financial statements separately.

German legislature enacts law on fair consumer contracts

A law on fair consumer contracts was passed in August 2021. The individual provisions of this law will enter into force in stages on 1 October 2021, 1 March 2022 and 1 July 2022. In particular, the law includes the obligation to implement a "cancellation button" from 1 July 2022 so that consumers have an easy-to-reach online channel for cancelling the contract.

IT Security Act 2.0 enters into force

In May 2021, the „Second Act to Increase the Security of Information Technology Systems“ (Zweites Gesetz zur Erhöhung der Sicherheit informationstechnischer Systeme) entered into force. It builds on the first act from 2015 and aims to better protect public structures from cyber attacks. This is especially true for critical infrastructures (KRITIS), such as telecommunications. As a result, certain components and manufacturers could be banned or excluded. On 23 August 2021, the BNetzA published a „General order for the definition of the catalogue of security requirements for the operation of telecommunications and data processing systems and for the processing of personal data (§ 109 paragraph 6 of the Telecommunications Act, „Security Catalogue“). The Security Catalogue contains a list of critical functions; this list is the basis for a prohibition or exclusion. Due to unclear wording in the list and the resulting legal uncertainty, the Telefónica Deutschland Group has filed an objection, which has not been decided on so far. On 7 October 2021, the Federal Ministry of the Interior and Community issued a „General order on the details of the minimum requirements for the guarantee declaration (prohibition of the use of critical components) for the telecommunications sector to be determined on the basis of the act covering the Federal Office for Information Security (BSI)“. The aim is to establish minimum requirements for manufacturers of critical components in warranty statements made to operators of critical infrastructure. Since there are no clear definitions, the Telefónica Deutschland Group has brought an action against this general order, which has not been decided on so far.

Discontinuation of data retention by BNetzA

In 2017, BNetzA published a notification according to which it will refrain from issuing ordinances and taking other measures to enforce the retention obligations in respect of all companies required to do so until the lawfulness of the retention obligations had been legally clarified. For this reason, Telefónica Deutschland Group temporarily discontinued the retention in 2017. The legally binding clarification of the legality of the storage obligation also continued in 2021.

Access and price regulation

MTR and FTR – delegated act enters into force

The EU Commission's delegated act entered into force on 1 July 2021 and the approved charges have been amended accordingly. It provides for the application of a glide path for the MTR (Mobile Termination Rate), according to which a charge of EUR 0.70 cents per minute will apply from 1 July 2021, decreasing to EUR 0.55 cents per minute from 1 January 2022, to EUR 0.40 cents per minute from 1 January 2023 and to EUR 0.20 cents per minute from 1 January 2024. FTRs (Fixed Termination Rate) have been subject to a charge of EUR 0.07 cents per minute since 1 July 2021. These charges will apply to all German providers of these services. In some cases, different charges apply to providers in other European countries.

EU Roaming Regulation to be revised

In November 2021, the EU Commission, the EU Council and the EU Parliament agreed on a first draft of a new version of the EU Roaming Regulation, which will take effect from 1 July 2022. Additional rounds of voting are still pending. The current principle of "roam-like-at-home" will be extended until 2032 and additional benefits and protections for consumers will be introduced. In all probability, the key innovations will be the introduction of Quality of Service (QoS), improved, free access to emergency call services, the obligation to provide information about and protections against additional charges abroad, such as for service numbers or in aeroplanes, as well as the reduction of wholesale charges. There are also plans to ask the EU Commission to look into measures on intra-EU calls, i.e. calls from the home country to an EU member state.

BNetzA continues regulatory procedures on fibre-optic infra-structures

The BNetzA has identified the existence of significant market power of Telekom Deutschland GmbH on markets 3a (= market for wholesale access provided locally at a fixed location) and 3b (= market for wholesale mass market products provided centrally at a fixed location) and still deems both wholesale markets to be in need of regulation. This applies to both the copper network of Telekom Deutschland GmbH and the newly constructed fibre optic network for the implementation of FTTH.

For market 3a, the BNetzA has determined that Telekom Deutschland GmbH has nationwide market dominance for ADSL, VDSL and FTTH. On 11 October 2021, the BNetzA published a draft decision on the framework conditions under which charges and other access conditions will be controlled by the BNetzA in future. This relates both to access to the local loop in the copper network, the so called "last mile", and also to virtual unbundled local access (VULA) to the copper and fibre connections of Telekom Deutschland GmbH. The Telefónica Deutschland Group submitted comments on the draft decision within the specified time limit.

With respect to market 3b, the BNetzA published a draft decision in the third quarter of 2020 in which Telekom Deutschland GmbH was identified as having market dominance only in cities with populations of more than 60,000. A draft decision on the type of regulation of fees and other access conditions for wholesale products in market 3b resulting from this will probably not be published until 2022.

Conclusion of an agreement with Telekom Deutschland GmbH for long-term access to xDSL and FTTH accesses

On 1 October 2020, the Telefónica Deutschland Group concluded binding long-term preliminary agreements with Telekom Deutschland GmbH, defining future access entitlements and corresponding conditions for access to xDSL and FTTH accesses. Most of the agreed conditions are subject to the BNetzA not prohibiting the fee agreements. No prohibition has been issued with regard to the agreed xDSL conditions. A review of the FTTH conditions is still ongoing. There are currently no indications that the BNetzA might decide to prohibit the charges at a later date.

Overview of the Financial Year 2021

Operational Performance

In financial year 2021, the Telefónica Deutschland Group delivered sustained solid operational and financial performance in a rational, while dynamic market environment, underpinned by the focused execution of the company's 'Investment for Growth' programme. As the German economy recovered and with the easing of the COVID-19 measures taken in the wake of the COVID-19 pandemic from the second quarter onwards, the gradual reopening of the O₂ shops was enabled. This reinforced Telefónica Deutschland Group's operational performance, with the O₂ Free tariff portfolio as a key driver.

In total, the Telefónica Deutschland Group recorded 1,526 thousand postpaid net adds (1,043 thousand in 2020) and 203 thousand M2M net adds (218 thousand in 2020) in the financial year owing to the positive development of the retail customer business in combination with a robust partner business. In the prepaid market, the trend of prepaid to postpaid migration continued unchanged. Against this backdrop, the Telefónica Deutschland Group recorded a decline of prepaid accesses by 310 thousand in 2021 (-813 thousand in the previous year).

Overall, the number of mobile accesses grew by 3.2% in 2021 and amounted to 45.7 million at the end of the year. The strongest growth driver (+6.5% year-on-year) was mobile postpaid excluding M2M, which recorded 25.1 million accesses as of 31 December 2021 (54.9% of total mobile base, up 1.7 percentage points year-on-year). Churn rates remained at historic lows, with the O₂ brand's implied annualised churn rate improving to 11.1% from 13.1% in 2020. This is confirmation of our sustained quality improvements and a multi-award winning customer experience on the O₂ network. The number of M2M connections totalled 1.6 million as of 31 December 2021, an increase of 14.4% compared to the prior year. In contrast, mobile accesses in the mobile prepaid segment declined by 1.6% year-on-year to 19.0 million, which reflects the ongoing prepaid-to-postpaid migration trend in the market.

Own brand postpaid ARPU grew 0.2% year-on-year in financial year 2021 despite the impact of travel restrictions on roaming revenues and the mobile termination rate cut brought forward to 1 July 2021.

Fixed broadband business returned to growth with the reopening of the O₂ shops while recording a total of +1 thousand net adds (+55 thousand in 2020) in financial year 2021. Accordingly, the fixed-line customer base remained largely stable at a total of 2.3 million customers (+0.1% year-on-year). Telefónica Deutschland Group also registered continued strong demand for mobile-based fixed-mobile substitution (FMS) products in its O₂ my Home portfolio.

Financial Performance

The Telefónica Deutschland Group's continued focus on profitable growth also resulted in a strong financial performance.

Revenues in the 2021 financial year were EUR 7,765 million, an increase of 3.1% year-on-year and thus in line with the expectations that had been raised in the first half-year. This development was driven in particular by the continued growth of mobile service revenues, which increased by 3.5% year-on-year despite the existence of a few remaining roaming restrictions and some pressure from the reduction in mobile termination rates, which has been brought forward to 1 July 2021. High-quality mobile phones remained popular, and handset sales rose by 1.9% despite some supply chain bottlenecks. Fixed broadband posted growth of 3.6% compared to the previous year.

Operating income before depreciation and amortisation (OIBDA), adjusted for exceptional effects came to EUR 2,411 million in the reporting year, +3.9% compared to the previous year (EUR 2,319 million) and thus remained within the expectations of the outlook upgraded twice. The adjusted OIBDA margin of 31.0% (+0.3 percentage points year-on-year) reflects operating momentum combined with the revenue mix and continued improvements in operating efficiency. This was partly offset by the EU/non-EU roaming mix and some inflation-related cost pressure, among other factors.

Exceptional effects in financial year 2021 totalled EUR 243 million (previous year: EUR 364 million) and mainly included the net gain on disposal of EUR 262 million from the second tranche of the sale of major parts of the business operations of the rooftop sites to Telxius Telecom, S.A. (>Note 7 DISPOSAL GROUPS<) as well as other income from the release of provisions in connection with M&A in the amount of EUR 3 million. This was offset by restructuring expenses of EUR 22 million (previous year: EUR 38 million). In the previous year, the sale of spectrum licences also resulted in net losses of EUR 5 million.

Capital Expenditure

Capital expenditure (CapEx) amounted to EUR 1,284 million in 2021 (previous year: EUR 1,094 million) with an investment ratio of 16.5% (previous year: 14.5%). The Telefónica Deutschland Group executed its three-year (2020 to 2022) network-focused Investment for Growth programme according to plan. The focus of investment was, as expected, in the second year, with the Telefónica Deutschland Group targeting the capital expenditure efficiency. The O₂ 5G network reached around 30% population coverage in Germany by the end of 2021. The company is rapidly scaling up 5G with the goal of nationwide coverage by 2025.

T 01

OVERVIEW OF THE FINANCIAL YEAR 2021

	Actual 2020 (in EUR million)	Outlook for 2021 ²⁰ (year-on-year)	Updated outlook for 2021 ²⁰ (year-on-year)	Financial year 2021 (year-on-year in %)	Assessment (compared to the original outlook)
Revenues	7,532	Unchanged to slightly positive year-on-year	Slightly positive year-on-year	EUR 7,765 million (+3.1%)	Better than expected
OIBDA adjusted for exceptional effects	2,319	Largely unchanged to slightly positive year-on-year	Low mid single-digit percentage growth	EUR 2,411 million (+3.9%)	Better than expected
Investment ratio (CapEx/Sales ratio)	14.5%	17 – 18%	<17 – 18%	16.5%	Better than expected

Sale of major parts of the business operations of the rooftop sites

In the 2020 financial year, the Telefónica Deutschland Group with Telxius Telecom, S.A. ("Telxius"), an affiliated company of the Telefónica, S.A. Group, the sale of major parts of the business operations of the rooftop sites.

This includes the spin-off and sale of a large part of its passive infrastructure on 10,113 mobile communications sites, consisting of 10,037 rooftop sites and 76 tower sites including the related lease agreements, associated assets and liabilities, know-how and other legal arrangements, at a nominal purchase price of EUR 1.5 billion. Due to the transfer of important processes, this involves the sale of a business as defined by IFRS 3. The sale of the business was recorded under IFRS 10.

The active radio technology at the transferred stations remains the property of the Telefónica Deutschland Group and will continue to be used by the company, unchanged, for operation of the mobile communications network.

The transaction of two steps: on 19 August 2020, around 60% of the sites (5,975 rooftop sites and 58 tower sites) were spun off to Telefónica Germany Mobilfunk Standortgesellschaft mbH ("TGMS"), a company newly in the first half of 2020. The shares in this company were transferred to Telxius on 1 September 2020. TGMS was then renamed Telxius Towers Erste GmbH. In December 2020, this company was merged into Telxius Towers Germany GmbH ("Telxius Germany").

With effect from 1 June 2021, ATC Germany Holdings GmbH has acquired all the shares in Telxius Germany; as a result, the latter was renamed ATC Germany Munich GmbH.

On 22 July 2021, a further approximately 40% of the sites (4,062 rooftop sites and 18 tower sites) were spun off to Telefónica Germany Zweite Mobilfunk Standortgesellschaft mbH ("TGZMS"), a company that was established in financial year 2020. The shares in this company were transferred to Telxius with effect from 1 August 2021 (^{->Note 7 DISPOSAL GROUPS<-}). Telxius then transferred all the shares in TGZMS to ATC Germany Holdings GmbH and the company was renamed ATC Germany Munich Zweite GmbH. ATC Germany Munich Zweite GmbH was merged into ATC Germany Munich GmbH with effect from 19 November 2021. With the specific selection of the individual transfer locations in June 2021, the associated assets and liabilities were identified individually. The assets and liabilities to be transferred were therefore reported as „held for sale“ in accordance with IFRS 5 during the course of the year, up to the date of sale on 1 August 2021.

The purchase price receivable including purchase price allocation from the second transaction step amounts to EUR 632 million. The amount of EUR 519 million was already paid within the reporting period. The remaining purchase price receivable of EUR 113 million is due for payment in 2022 with a share of EUR 16 million. A further EUR 97 million will be due for payment in 2025.

With the transfer of the TGZMS to Telxius and the subsequent transfer to ATC Germany Holdings GmbH, the Telefónica Deutschland Group will lease areas of the transferred passive infrastructure from ATC Germany Munich GmbH for the installation and operation of its active radio technology.

The entire transaction was approved by the competent authorities in the previous year.

Business development is further detailed in the following sections.

²⁰ The financial outlook originally issued for the financial year 2021 in the Annual Report dated 3 March 2021 for revenues and OIBDA adjusted for exceptional effects, was raised in the context of the H1 2021 results and OIBDA adjusted for exceptional effects, was raised again in the context of the Q3 results. The expectations for the investment ratio were also adjusted as a result.